복사전달 특강

Special Topics in Radiative Transfer

Lecture 8

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updated on 04/20

Based on: Chapter 6 of Noebauer & Sim (Living Reviews in Computational Astrophysics)

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Chapter 6: Propagation of Quanta

The core computational loop of any MCRT simulation

- After initialization, packets are propagated through the domain.
- This constitutes the bulk of MCRT runtime.

1. Draw τ

Sample optical depth:

$$\tau = -\ln \xi$$

2. Locate interaction

Invert $\int_0^l dl' \chi(l') = \tau$

to find distance l.

Here, $\chi(l')$ is the opacity.

3. Compare distances

distance l vs. cell boundary,
vs. time step end

→ take minimum,
assign the shortest distance to l.

4. Move packet

- Advance by l
- Update time stamp $t = t + l/c$

5. Perform event

- Scattering
- Absorption, or
- Cell crossing

6. Termination?

- Escape through one of the domain boundaries
- Absorption interaction
- Reaching the end of a pre-defined time interval.

Fig. 2: Packet Propagation Algorithm

The two-loop structure of the MCRT propagation algorithm

Outer Loop: Packet Loop

- Iterates over all MC packets in the population
- Each packet processed through propagation loop
- Returns to outer loop once packet terminates

Inner Loop: Propagation Loop

- Moves packet step by step through domain
- Each step: draw τ → invert → move → event
- Loop continues until termination condition

Update Estimators

- At each step, packet properties are recorded to reconstruct radiation field quantities (energy density, flux, etc.) from the ensemble. See Sect. 9.

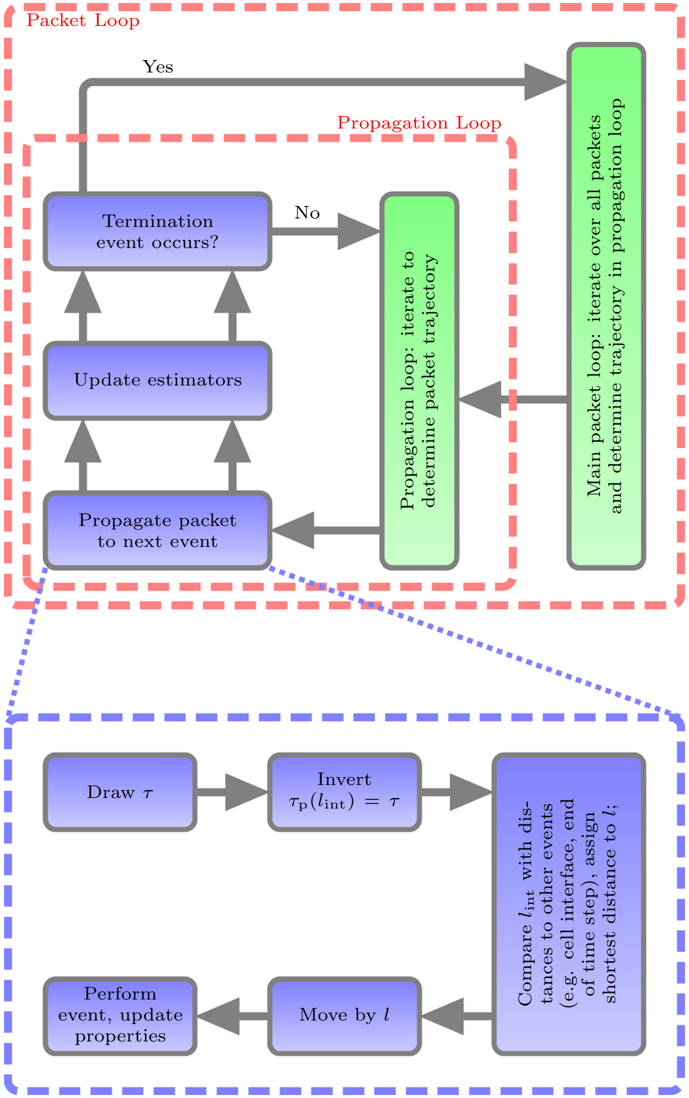


Illustration of the packet process.

- At its core, the main processing loop works through the entire packet population, propagating each packet individually. In this loop, each packet is moved through the domain until a termination event occurs. Depending on the specific MCRT application, this may for example be an absorption interaction, escaping through the domain boundaries or reaching the end of a pre-defined time interval.
- The instruction “update estimators” refers to all activities related to recording packet properties that are used to reconstruct physical information about the radiation field from the packet ensemble.

6.2 Absorption: Discrete vs. Continuous

Two complementary approaches to handling absorptive opacity

Discrete Interaction Events

- Packet propagates until τ threshold reached
- Absorption terminates packet at that point
- In radiative equilibrium (RE; Lucy 1999a): immediately re-emitted
- Required for line interactions where absorption ↔ re-emission must be linked
- Used in: Abbott & Lucy 1985; Long & Knigge 2002

Continuous Weight Degradation

- Weight decays along flight path: $w(l) = w_0 \cdot \exp(-\tau_a)$, where $\tau_a(l) = \int_0^l dl' \chi_a(l')$ is the optical depth due to the absorption opacity χ_a.
- Smoother → lower MC noise for small opacity
- Disconnects absorption from re-emission; new packets must be injected separately
- Suits pure-absorption problems
- Hybrid schemes common (continuous attenuation + discrete line interaction or scattering)

This review focuses primarily on the discrete interaction approach; most principles also apply when weight degradation is used for continuum opacity.

6.3 Numerical Discretization & Material Properties

Implementing MCRT on a computational grid

- Real applications require **a spatial grid dividing the domain into cells.**
  Material properties (density, temperature, velocity) are represented discretely per cell.

Material State per Cell

- Sets local opacity → rate of optical depth accumulation $\tau_{\mathrm{p}}(l) = \int_0^l dl' \chi(l')$.
- Also dictates re-emission characteristics in scattering events.

Cell Crossing Events

- When a packet crosses a cell boundary, opacities are recalculated.
- τ may be redrawn (exploiting the memoryless property of the exponential distribution).

Opacity Integration

- Within a constant-opacity cell: $l = \tau / \kappa$ (trivial inversion of $\tau_{\mathrm{p}}(l) = \int_0^l dl' \chi(l')$).
- More complex for fast flows where opacity is frequency/direction dependent (Sect. 8).

Grid Types

- Cartesian, spherical, cylindrical, or unstructured — chosen based on problem geometry.
  Adaptive Mesh Refinement (AMR) is also commonly used.
- Interpolation between cells is possible but adds computational cost.

6.4 Absorption & Scattering Interactions

Handling multiple opacity sources probabilistically

- Total opacity: $\kappa_{\text{tot}} = \kappa_{\text{s}} + \kappa_{\text{a}}$
- Interaction distance: $l = \tau / \kappa_{\text{tot}}$

Decision at the interaction point:

Draw $\xi \in$ [0, 1)

If $\xi \leq \kappa_{\text{s}}/(\kappa_{\text{s}} + \kappa_{\text{a}})$ → SCATTER

Else → ABSORB (remove packet, or re-emit in RE applications)

After Scattering (isotropic coherent example):

- New direction drawn: $\mu = 2\xi_1 - 1,\ \ \phi = 2\pi\xi_2$
- Frequency unchanged (coherent); new τ drawn; accumulated optical depth reset
- Multiple scatterings handled naturally — a major strength of MCRT over deterministic methods!
- Easily extended: anisotropic scattering simply changes the directional sampling rule

6.5 Propagation Example: Homogeneous Sphere

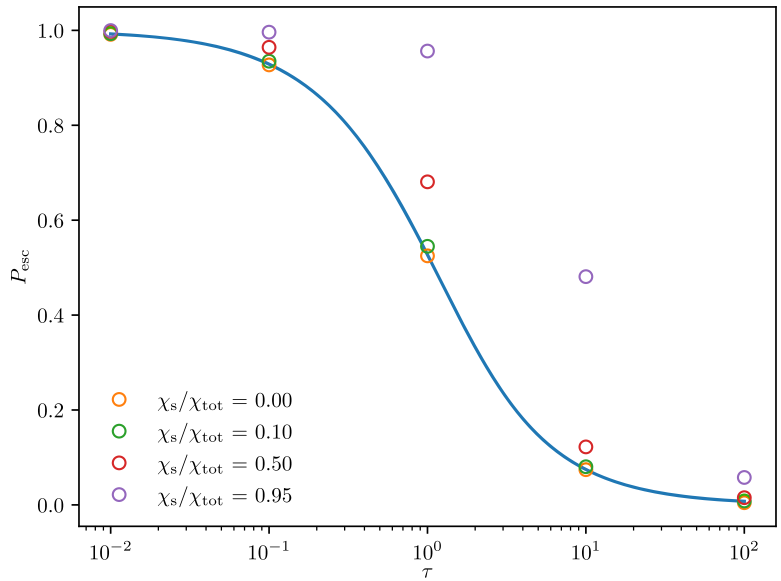
MCRT test case — escape probability from a uniform sphere

- Test problem: photon escape from a sphere of uniform density and emissivity.
- Analytic solution available for pure absorption (see Appendix A.1).
- Packets initialized at $r = R\xi^{1/3}$, direction $\mu = 2\xi - 1$.

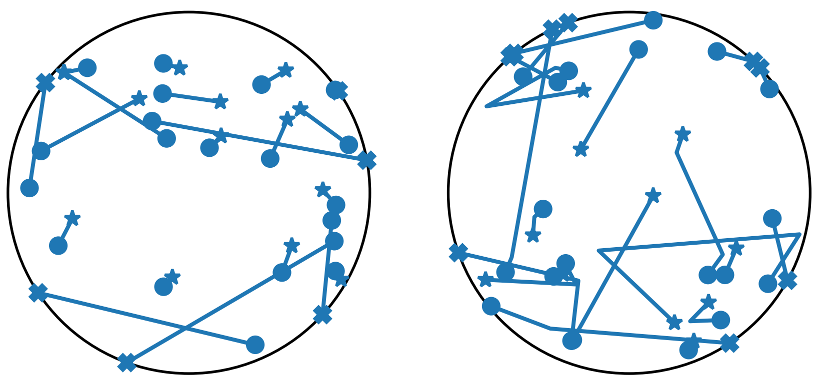
Python Code

Escape probability from a uniform sphere

https://github.com/unoebauer/mcrtreview-tools/esc_probability_test



Escape probability $P_{\rm esc}$ vs. total optical depth τ for different scattering fractions. Solid line = analytic solution (pure absorption). MCRT agrees well.



Sample packet trajectories for $\tau = 2$.
Left: $\kappa_s = 0$ (pure absorption).
Right: $\kappa_s = \kappa_a$. ● = start, ★ = absorbed, × = escaped.

6.6 Time-Dependent MCRT

Extending steady-state MCRT to evolving radiation fields

- Time-dependence requires minimal changes: each packet carries an internal clock t, updated as $\Delta t = l/c$ after each propagation step.

Time Stamp

- Each packet: t initialized from emission process.
- Updated by $\Delta t = l/c$.

Time Grid

- Simulation subdivided into time steps t_n.
- Material state constant per step.

Interruption Rule

- Packet reaching t_{n+1} is paused; state saved.
- Resumes in next cycle.

Initialization in time-dependent problems:

- Initial radiation field packets: t = starting time of current simulation phase
- Source packets (continuous emitter): t drawn uniformly over current time step duration
- Material state updated between time steps (coupling with hydrodynamics → Sect. 11)
- Problems arising from very rapid material state changes: addressed in Sect. 10

Memoryless property of the exponential distribution

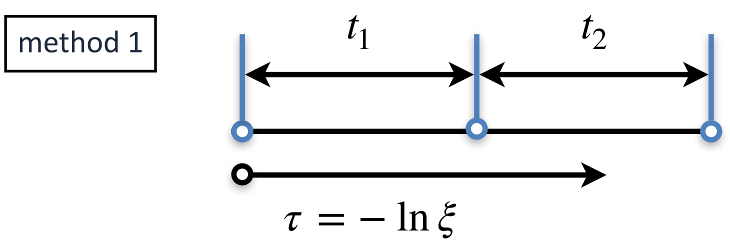
Finding where the photon packet interacts with medium.

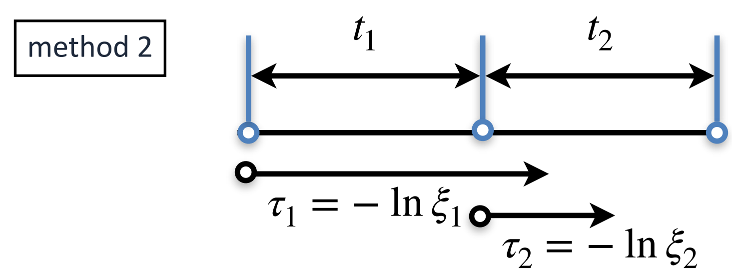
Method 1

- A single sampled optical depth is used to determine the location across multiple cells.

Method 2

- Whenever the photon crosses a cell boundary, the optical depth is sampled again and compared with the distance the photon travels through the cell, until the sampled optical depth becomes less than that distance.





In the left figure, the optical depth is determined using Method 1, and the probability of the packet interacting within the second cell is:

$$P(t_1 < \tau < t_1 + t_2) = \int_{t_1}^{t_1+t_2} e^{-\tau} d\tau = e^{-t_1} - e^{-(t_1+t_2)}$$

In the right figure, when the packet trajectory is simulated using Method 2, the packet does not interact within the first cell on the first draw, but interacts within the second cell after the trajectory is reset and the optical depth is resampled at the boundary between Cells 1 and 2. The probability of interaction within Cell 2 after the two draws is then:

$$P(\tau > t_1)P(\tau < t_2) = e^{-t_1}\left(1 - e^{-t_2}\right)$$

Therefore, the two probabilities above are found to be identical.

The review by Noebauer & Sim refers to this property as the **infinite divisibility**; however, this terminology appears to be inappropriate.

Memoryless property of the exponential distribution

- The probability of an event occurring in the future is independent of how much time has already elapsed.
  Mathematically, it is defined by

  $$P(X > t_1 + t_2 | X > t_1) = P(X > t_2) \text{ for all } t_1.$$

- This means the probability of waiting an additional t_2 time, given we've already waited t_1, is simply the probability of waiting t_1 from the beginning.
  For instance, if a lightbulb's lifespan is exponentially distributed, a bulb that has already worked for 100 hours is just as likely to last another 50 hours as a brand-new bulb.

  Proof:

  $$P(X > t_1 + t_2 | X > t_1) = \frac{P(X > t_1 + t_2)}{P(X > t_1)} = \frac{e^{-\lambda(t_1+t_2)}}{e^{-\lambda t_1}} = e^{-\lambda t_2} = P(X > t_2)$$

- The two sampling methods 1 and 2 are equivalent by the memoryless property:

  Proof:

  $$\begin{aligned}
  P(t_1 < \tau < t_1 + t_2) &= P(\tau > t_1) - P(\tau > t_1 + t_2) \\
  &= P(\tau > t_1) - P(\tau > t_1 + t_2 | \tau > t_1) P(\tau > t_1) && \Leftarrow P(A, B) = P(A | B) P(B) \\
  &= P(\tau > t_1) \left[1 - P(\tau > t_1 + t_2 | \tau > t_1)\right] \\
  &= P(\tau > t_1) \left[1 - P(\tau > t_2)\right] && \Leftarrow \text{memoryless property} \\
  &= P(\tau > t_1) P(\tau < t_2)
  \end{aligned}$$

Summary of Chapters 5 & 6

Ch. 5: Two Discretizations

Photon packets conserve photon number.
Energy packets conserve energy.

Choice depends on scientific goal — neither universally superior.

Ch. 5: Initialization

Packets draw position, direction, and frequency from appropriate distributions
(Planck for frequency; isotropic or flux-based for direction).

Ch. 6: Propagation Loop

Draw $\tau = -\ln \xi$ → locate interaction → move to minimum-distance event → scatter or absorb
→ repeat until termination condition.

Ch. 6: Scattering Strength

Multiple, anisotropic scattering is handled naturally in MCRT — a major advantage over deterministic RT solvers with non-local coupling.

Example: Homogeneous Sphere

MCRT test case — escape probability from a uniform sphere

- We consider a homogeneous sphere of radius R composed of material with constant absorption coefficient α and emissivity ε. The escape probability as a function of optical depth ($\tau = \alpha R$) is given by

$$p_{\rm esc}(\tau) = \frac{3}{4\tau}\left[1 - \frac{1}{2\tau^2} + \left(\frac{1}{\tau} + \frac{1}{2\tau^2}\right)\exp\left(-2\tau\right)\right]$$

(Osterbrock 1974, Appendix 2)

  The probability is the emergent flux at the surface of the sphere, relative to the expected flux in the absence of absorption. This formula gives the escape probability for the case of no scattering (albedo $a = 0$).

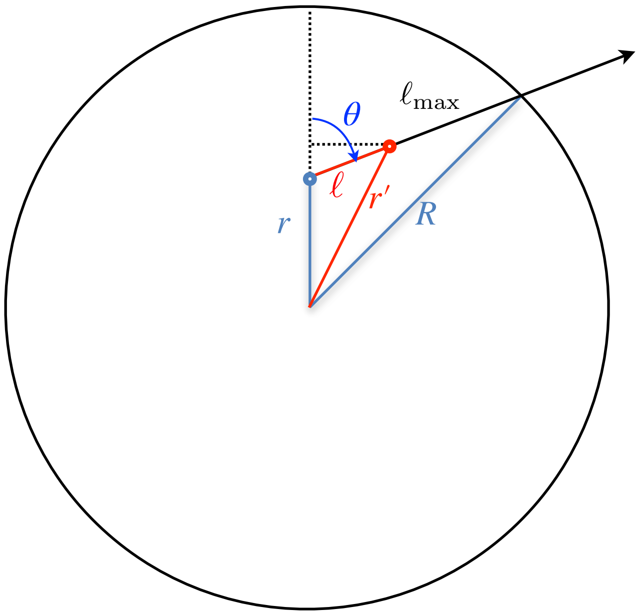
- Monte Carlo Simulation:

  The python code calculate the escape probability for an arbitrary scattering albedo $a = \alpha_{\rm scat}/\left(\alpha_{\rm scat} + \alpha_{\rm abs}\right) > 0$.

  This code assumes isotropic scattering and therefore spherical symmetry, so it is much simpler than what we discussed.

- Can we write a more general python code that can be extended to handle anisotropic scattering?

• In the spherically symmetric case, the simulation can be simplified.



cosine law:

$$R = \sqrt{\ell_{\max}^2 + r^2 - 2\ell_{\max} r \cos(\pi - \theta)}$$

Here, $\ell_{\max}$ is the distance to the sphere boundary.

$$R^2 = \ell_{\max}^2 + r^2 + 2\ell_{\max} r \cos(\theta)$$

$$\ell_{\max} = -r\mu + \sqrt{R^2 - r^2(1-\mu^2)}$$

Now, the radial distance r' of the scattering point is given by

$$(\ell\cos\theta + r)^2 + (\ell\sin\theta)^2 = r'^2$$

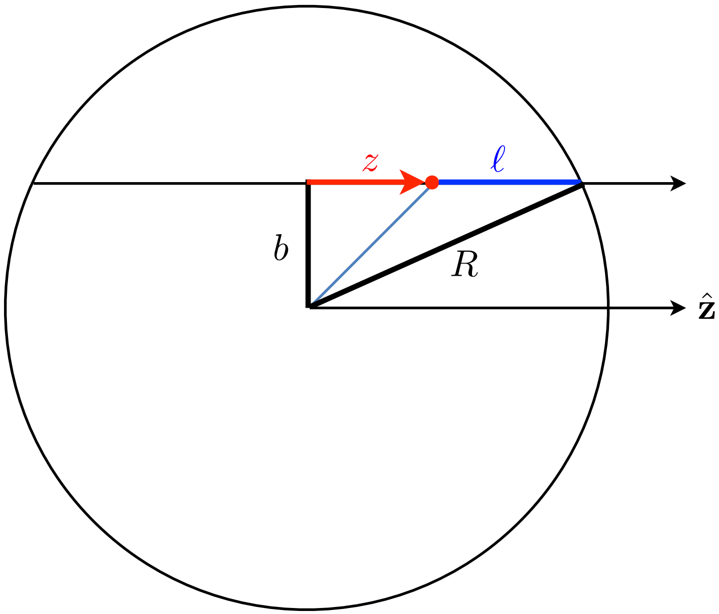
$$r' = \sqrt{\ell^2 + r^2 + 2\ell r\cos\theta}$$

• Escape probability for the pure absorption case:

$$p_{\rm esc} = \frac{\displaystyle\int d\Omega \int dV \varepsilon \cdot e^{-\tau(\mathbf{r},\mathbf{\Omega})}}{\displaystyle\int d\Omega \int dV \varepsilon}$$

Because of the symmetry, this value should be equal to the ratio obtained for a specific direction, for instance, $\hat{\mathbf{z}}$:

$$p_{\rm esc} = \frac{\displaystyle\int dV e^{-\tau(\mathbf{r},\hat{\mathbf{z}})}}{\displaystyle\int dV}$$



$$dV = 2\pi b db \cdot dz$$

$$0 \leqslant b \leqslant R$$

$$z_{\rm min} \leqslant z \leqslant z_{\rm max} \quad \text{where } z_{\rm max} = \sqrt{R^2 - b^2} = -z_{\rm min}$$

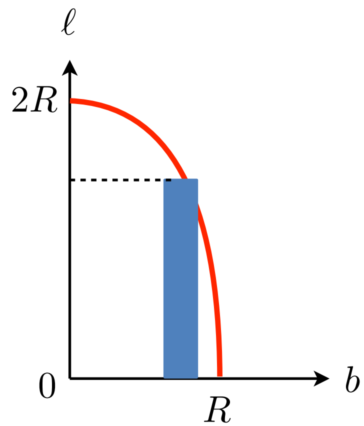
$$p_{\rm esc} = \frac{2\pi}{(4\pi/3)R^3} \int_0^R b db \int_{-z_{\rm max}}^{z_{\rm max}} dz e^{-\alpha\ell}$$

Here, we note that $z + \ell = z_{\rm max}$

$$dz = -d\ell \quad \text{and} \quad 0 \leqslant \ell \leqslant 2\sqrt{R^2 - b^2}$$

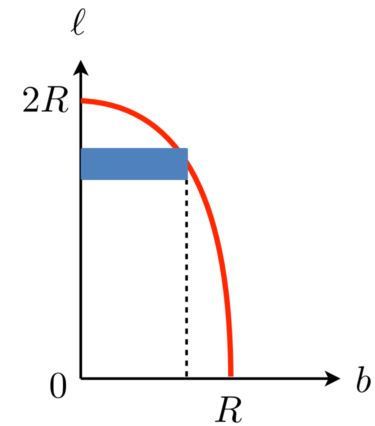
Therefore,

$$p_{\rm esc} = \frac{3}{2R^3} \int_0^R b db \int_0^{2\sqrt{R^2-b^2}} d\ell e^{-\alpha\ell}$$



equation of the red curve

$$\ell^2 = 4\left(R^2 - b^2\right)$$



Integrate over ℓ for a given b, and then integrate over b

Integrate over b for a given ℓ, and then integrate over ℓ

Interchanging the order of integral:

$$\begin{aligned} p_{\rm esc} &= \frac{3}{2R^3}\int_0^{2R} d\ell \int_0^{\sqrt{R^2-\ell^2/4}} db\, b e^{-\alpha\ell} \\ &= \frac{3}{4R^3}\int_0^{2R} d\ell e^{-\alpha\ell}\left(R^2 - \frac{\ell^2}{4}\right) \\ &= \frac{3}{4}\int_0^2 du\left(1 - \frac{u^2}{4}\right)e^{-\tau u} \qquad \Leftarrow \ \text{let } u = \frac{\ell}{R} \text{ and } \tau = \alpha R \end{aligned}$$

Integral limit for ℓ for a given b:

$$0 \leqslant \ell \leqslant \ell_{\rm max} = 2\sqrt{R^2 - b^2}$$

Integral limit for b for a given ℓ:

$$0 \leqslant b \leqslant b_{\rm max} = \sqrt{R^2 - \ell^2/4}$$

Final result is then obtained as:

$$p_{\rm esc}(\tau) = \frac{3}{4\tau}\left[1 - \frac{1}{2\tau^2} + \left(\frac{1}{\tau} + \frac{1}{2\tau^2}\right)\exp\left(-2\tau\right)\right]$$

Example: Thomson scattering

Thomson scattering by an electron gas at temperature T

The probability density function for the velocity component u of a scattering electron parallel to the incoming photon is

$$P(v) = (\text{cross section}) \cdot \left(\frac{m_e}{2\pi k_\mathrm{B} T}\right)^{1/2} \exp\left(-\frac{m_e v^2}{2k_\mathrm{B}T}\right) = \frac{1}{\sqrt{\pi v_\mathrm{th}^2}} \exp\left(-\frac{v^2}{v_\mathrm{th}^2}\right)$$

where $v_\mathrm{th} = \sqrt{\dfrac{m_e}{2k_\mathrm{B}T}}$

Therefore, the velocity component can be drawn independently of the photon frequency.

(1) Initial photon location: $(x, y, z) = (0,0,0)$, and start from an initial normalized frequency $x \equiv \nu/\nu_0 = 1$.

(2) Choose an initial photon direction vector ($\mathbf{\Omega}_i$).

(3) Choose an optical depth, where the photon is scattered: $\tau = -\ln \xi \quad \leftarrow \xi \in U(0,1)$

(4) Choose a velocity component v_z parallel to the incoming photon from the above PDF.

(5) Choose two velocity components v_x and v_y perpendicular to the photon direction.

(6) Transform the frequency from LF to CMF: $\nu = \nu_i - \dfrac{v_z}{c}\nu_i$

Then, $x = x_i - \dfrac{v_z}{c} x_i$.

Choose three normal Gaussian random numbers:

$$\xi_1, \xi_2, \xi_3 \sim \mathrm{Gauss}(0,1).$$

Then,

$$v_z \sim v_{\mathrm{th}}/\sqrt{2} \cdot \xi_1$$

$$v_x \sim v_{\mathrm{th}}/\sqrt{2} \cdot \xi_2$$

$$v_y \sim v_{\mathrm{th}}/\sqrt{2} \cdot \xi_3$$

(7) Choose scattering angles μ and ϕ in the CMF.

(8) Update the photon propagation direction: $\mathbf{\Omega}_i \rightarrow \mathbf{\Omega}_f$ using μ and ϕ

(9) Transform the frequency from the CMF to LF: $\nu_f = \nu + \dfrac{\mathbf{v} \cdot \mathbf{\Omega}_f}{c}\nu$

In terms of the dimensionless variable:

$$x_f = x + \frac{\mathbf{v} \cdot \mathbf{\Omega}_f}{c} x$$

(10) Goto step (3) until the photon escape the sphere.